DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: Purple Company – Beauty Products E-Commerce Order Management System

1. Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.

1.1 COUNT ()

Find the total number of orders.

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM PurpleOrder;

TOTAL_ORDERS
------------
           5
```

1.2 SUM ()

Find the total sales amount.

```
SQL> SELECT SUM(Quantity * Unit_Price) AS Total_Sales
  2  FROM PurpleOrderItem;

TOTAL_SALES
-----------
      13298
```

1.3 – AVG ()

To calculate the average sales amount in the Purple Ecommerce database using the AVG () function.

```
          799

SQL> SELECT AVG(Order_Total) AS Average_Order_Amount
  2  FROM (
  3      SELECT Order_ID,
  4             SUM(Quantity * Unit_Price) AS Order_Total
  5      FROM PurpleOrderItem
  6      GROUP BY Order_ID
  7  );

AVERAGE_ORDER_AMOUNT
--------------------
              3324.5
```

1.4 MIN ()

To find the minimum product price in the Purple Ecommerce database using the MIN () function.

```
SQL> SELECT MIN(Order_Total) AS Lowest_Order
  2  FROM (
  3      SELECT Order_ID,
  4             SUM(Quantity * Unit_Price) AS Order_Total
  5      FROM PurpleOrderItem
  6      GROUP BY Order_ID
  7  );

LOWEST_ORDER
------------
        1800
```

1.5 MAX ()

To find the maximum product price in the Purple Ecommerce database using the MAX () function.

```
SQL> SELECT MAX(Order_Total) AS Highest_Order
  2  FROM (
  3      SELECT Order_ID,
  4             SUM(Quantity * Unit_Price) AS Order_Total
  5      FROM PurpleOrderItem
  6      GROUP BY Order_ID
  7  );

HIGHEST_ORDER
-------------
         6098
```

2. Generate Total Sales Reports

A sales report can be generated using the SUM () function on the product quantity and unit price.

```
SQL> SELECT SUM(Quantity * Unit_Price) AS Total_Sales
  2  FROM PurpleOrderItem;

TOTAL_SALES
-----------
      13298
```

2.1 Customer-wise Sales Report

To find how much each customer has purchased:

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         SUM(OI.Quantity * OI.Unit_Price) AS Total_Purchase
  4  FROM PurpleCustomer C
  5  JOIN PurpleOrder O
  6  ON C.Customer_ID = O.Customer_ID
  7  JOIN PurpleOrderItem OI
  8  ON O.Order_ID = OI.Order_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
         1
Arun
         6098

         2
Kavin
         3000

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------

         3
Priya
         2400

         4
Anu
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
          1800
```

3. Find Top Customers Based on Purchase Amount

The SUM () function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         SUM(OI.Quantity * OI.Unit_Price) AS Total_Purchase
  4  FROM PurpleCustomer C
  5  JOIN PurpleOrder O
  6  ON C.Customer_ID = O.Customer_ID
  7  JOIN PurpleOrderItem OI
  8  ON O.Order_ID = OI.Order_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
          1
Arun
          6098

          2
Kavin
          3000

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------

          3
Priya
          2400

          4
Anu
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
          1800
```

4. Identify Best-Selling Products

The PurpleOrderItem table contains the quantity of each product ordered.

Therefore, we can use SUM(Quantity) to find the total quantity sold for each product.

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         SUM(OI.Quantity) AS Total_Quantity_Sold
  4  FROM PurpleProduct P
  5  JOIN PurpleOrderItem OI
  6  ON P.Product_ID = OI.Product_ID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
         3
Waterproof Eyeliner
                  3

         4
Glow Blush
                  2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

         1
Matte Lipstick
                  2

         2
Liquid Foundation
```

4.1 Product Sales Amount

Instead of only calculating quantity, we can calculate the total sales generated by each product.

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         SUM(OI.Quantity) AS Total_Quantity,
  4         SUM(OI.Quantity * OI.Unit_Price) AS Total_Sales
  5  FROM PurpleProduct P
  6  JOIN PurpleOrderItem OI
  7  ON P.Product_ID = OI.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Total_Sales DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------
         1
Matte Lipstick
             2        4598

         3
Waterproof Eyeliner
             3        3000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES
-------------- -----------

         4
Glow Blush
             2        2400

         5
Volume Mascara
```

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each beauty product category.

```
SQL> SELECT C.Category_ID,
  2         C.Category_Name,
  3         SUM(OI.Quantity * OI.Unit_Price) AS Category_Sales
  4  FROM PurpleCategory C
  5  JOIN PurpleProduct P
  6  ON C.Category_ID = P.Category_ID
  7  JOIN PurpleOrderItem OI
  8  ON P.Product_ID = OI.Product_ID
  9  GROUP BY C.Category_ID, C.Category_Name
 10  ORDER BY Category_Sales DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
CATEGORY_SALES
--------------
         1
Lipstick
         4598

         3
Eyeliner
         3000

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
CATEGORY_SALES
--------------

         4
Blush
         2400

         5
Mascara
```

6. Combined Business Sales Report

A complete sales report can combine customer, order, product, and category information.

```
SQL> SELECT C.Customer_Name,
  2         O.Order_ID,
  3         P.Product_Name,
  4         CAT.Category_Name,
  5         OI.Quantity,
  6         OI.Unit_Price,
  7         (OI.Quantity * OI.Unit_Price) AS Amount
  8  FROM PurpleCustomer C
  9  JOIN PurpleOrder O
 10  ON C.Customer_ID = O.Customer_ID
 11  JOIN PurpleOrderItem OI
 12  ON O.Order_ID = OI.Order_ID
 13  JOIN PurpleProduct P
 14  ON OI.Product_ID = P.Product_ID
 15  JOIN PurpleCategory CAT
 16  ON P.Category_ID = CAT.Category_ID
 17  ORDER BY O.Order_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE     AMOUNT
---------- ---------- ----------
Arun
         1
Matte Lipstick
```

```
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE     AMOUNT
---------- ---------- ----------
Lipstick
         2       2299       4598


CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE     AMOUNT
---------- ---------- ----------
Arun
         1
Liquid Foundation

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE     AMOUNT
---------- ---------- ----------
Foundation
         1       1500       1500
```

```
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Kavin
         2
Waterproof Eyeliner

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Eyeliner
         3       1000       3000


CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Priya
         3
Glow Blush
```

```
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Blush
         2       1200       2400


CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Anu
         4
Volume Mascara

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE      AMOUNT
---------- ---------- ----------
Mascara
         1       1800       1800
```